

Shubham Singh

Software Developer

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SUMMARY

- **Software Developer** with **5 years** of experience in designing and implementing innovative web applications, specializing in back-end technologies.
- Expertise in Java frameworks such as **Spring Boot, Spring MVC, Spring Security**, and Hibernate for building robust and scalable applications
- Proficient in front-end development using technologies like **React.js, Angular, and JavaScript/TypeScript** to create engaging and responsive user interfaces.
- Skilled in cloud technologies and containerization tools, including **AWS, Docker, and Kubernetes**, to enhance application deployment and scalability.
- Experienced with various databases (**MySQL, MongoDB, Oracle**) and adept at designing data models and implementing efficient data access using **JPA/Hibernate**.
- Familiar with **Agile methodologies**, participating in the entire software development lifecycle from requirements gathering to design, development, testing, and deployment.
- Proficient in CI/CD practices, utilizing tools like **Jenkins** to set up reliable and automated deployment pipelines for seamless application delivery.
- Strong documentation skills, including API documentation with tools like **Swagger**, to ensure maintainability and support.

SKILLS

Front-End Technologies:	HTML5, CSS3, Bootstrap, UI Material, JavaScript, Typescript, ES6, React JS, Redux, React Hooks, Angular, Prime Faces
Back-End Technologies:	Core Java, Java 8, JTEE, Python, C++, PHP, Go Lang, Spring Boot, JDMC, Servlets, Spring Web Services, REST API, Microservices, Hibernate ORM, Flutter
Methodology	SDLC, Waterfall, Agile
Frameworks	Spring Framework, Spring MVC, Spring Boot, Spring Cloud, Spring ORM, Spring Security, Spring Integration, Hibernate, JDBC, Struts, Microservices, JSP, Multi-threading
Build Tools & IDEs	Maven, Gradle, Jenkins, NetBeans, IntelliJ IDEA, Eclipse
Application Servers	Web Logic, Apache Tomcat
Cloud & Deployment	AWS, Docker
API Development	RESTful services, SOAP, WSDL, Jersey Framework
Databases	MySQL, MS SQL Server, SQL Plus, Oracle, DB2, MongoDB
CI/CD & Tools	Jenkins, Code Ship
Testing tools	Junit, Mockito, Selenium
Version Control/Tracking Tools	Git, GitHub, Jira, SVN
Operating Systems	Windows, LINUX, Mac OS

EXPERIENCE

Merck & Co.| Software Developer, CA

Aug 2023 - Present

- Led the transition of a legacy monolithic healthcare application to a microservices architecture using Spring Boot, resulting in a 40% improvement in transaction processing speed by decomposing the application into independent services, thereby enhancing scalability, and reducing system latency.
- Integrated SSL/TLS encryption and OAuth 2.0 authentication protocols to safeguard sensitive patient data, ensuring compliance with HIPAA standards and significantly mitigating the risk of data breaches while enhancing overall application security.
- Deployed Apache Kafka for real-time data streaming, enabling efficient handling of large volumes of healthcare transactions and ensuring low-latency data processing for timely and accurate updates critical to patient care.

- Conducted extensive optimization of MySQL database queries through advanced indexing strategies and query restructuring, achieving a 30% improvement in data retrieval performance essential for high-frequency transaction processing within the healthcare environment.
- using WSDL to define the service interface, allowing for structured and standardized communication between distributed systems, ensuring interoperability with other healthcare applications.
- Implemented event handling mechanisms using Spring Events to enable asynchronous communication between microservices, enhancing responsiveness and decoupling service interactions for improved system performance.
- Deployed Java-based microservices on Amazon EC2 instances, leveraging auto-scaling groups to ensure high availability and scalability in response to fluctuating healthcare data volumes.
- Utilized Java 8 features, including lambda expressions and the Stream API, to enhance code efficiency and readability in backend development, contributing to more maintainable code and improved application performance.
- Actively participated in sprint planning sessions, where user stories related to the migration to a microservices architecture were broken down and estimated, assisting the team in effective resource allocation, and maintaining focus on delivering high-priority features for healthcare applications.

Goldman Sachs | Software Developer, NJ

Jan 2019 – Jul 2022

- Designed and implemented a distributed system using Spring Boot and Apache Kafka, enhancing real-time data processing capabilities and reducing latency by 35%.
- Integrated **JMS** for asynchronous communication among distributed components, configuring message producers and consumers to ensure reliable message delivery.
- Developed reusable and modular React components to maintain a scalable codebase, employing functional components with hooks for effective state management and lifecycle handling.
- Architected a microservices framework to decompose the application into loosely coupled, independently deployable services, increasing fault tolerance and scalability.
- Leveraged Spring Data MongoDB for efficient database management, utilizing Mongo Repository for CRUD operations and implementing custom queries for optimized data access.
- Integrated advanced machine learning models using Python libraries (scikit-learn), resulting in a 15% improvement in credit risk assessment accuracy.
- Implemented multithreading techniques to enhance application performance, utilizing Java's Executor framework to manage thread pools, optimize resource usage, and ensure smooth execution of concurrent tasks.
- Executed horizontal scaling strategies to manage increased load by deploying additional instances of microservices, utilizing Kubernetes for container orchestration and resource management.
- Incorporated Jenkins into the CI/CD pipeline, automating build, test, and deployment processes to significantly minimize deployment errors and enhance delivery speed.
- Developed SOAP-based web services for secure and structured communication between distributed components, ensuring interoperability with legacy systems and adherence to established protocols.
- Hands-on experience in Google Cloud Platform (GCP), Azure at leveraging cloud services for scalable and resilient solutions.
- Generated comprehensive API documentation using Swagger and OpenAPI, providing interactive guides to help developers understand and test API endpoints efficiently.
- Employed Mockito for unit testing, creating mock objects for service and controller dependencies to ensure isolated testing that focuses on specific class logic without reliance on actual implementations.

EDUCATION

Master of Science in Data Analytics: San Jose State University.

Bachelor of Science in Computer Science and Engineering: State University of New York at Buffalo

CERTIFICATION AND COURSES

Excel (Beginners to Advanced)

AWS cloud practitioner

Courses - Data Structure, Object Oriented Programming in Java, Database Management, Operating System, Machine Learning, Algorithms